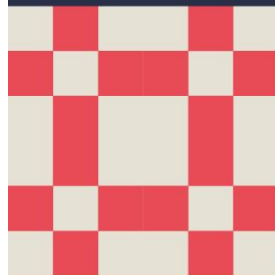


Moodle App Testing: Then and Now

Presented by Noel De Martin
Analyst Developer (Moodle HQ)



Together, we can
make the world a
better place



20
23



moodlemoot.org

#MootGlobal23

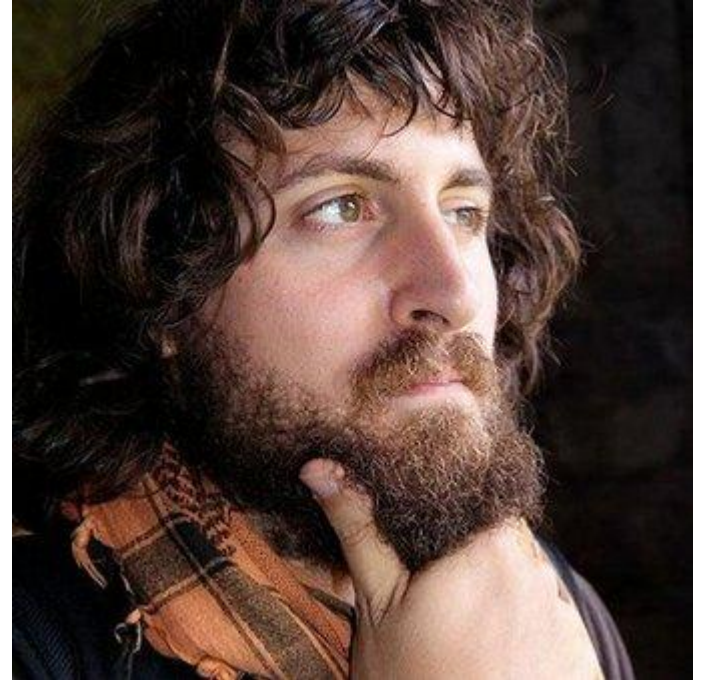
Maybe you met me
yesterday...



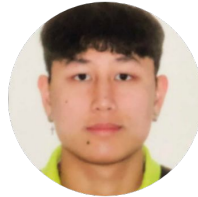


Hey! My name is Noel

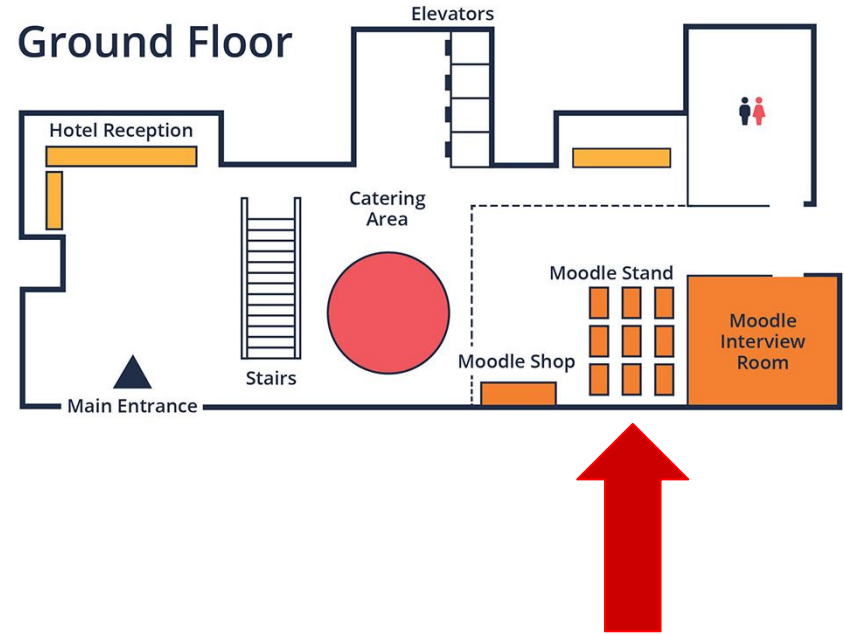
- Joined Moodle 4 years ago
(same time as the 2019 Global Moot)
- Developer working on the Moodle App
- First time presenting in a live Moot!



The Mobile Solutions Team



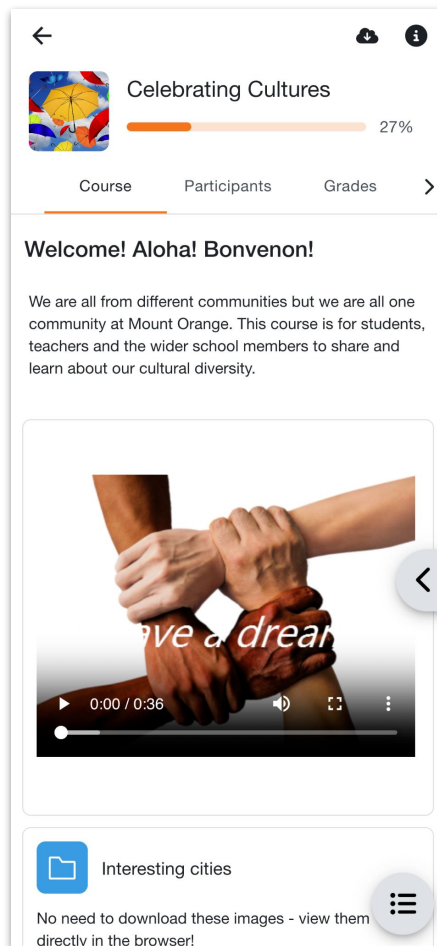
**Come visit us at the
Moodle Stand!**



The Moodle App

<https://github.com/moodlehq/moodleapp>

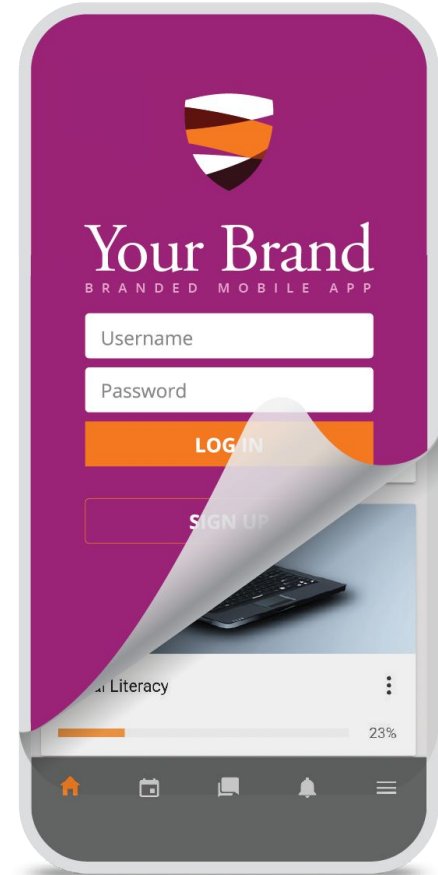
- For Android and Apple devices
- Students fully supported
- Works offline
- Supports plugins
- Push notifications
- Free to use
- Open source
- ≈8M active users per month

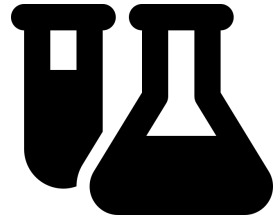


Premium services

- Pro/Premium Plans with additional features
<https://moodle.com/solutions/moodle-app/>
- Branded Moodle App
<https://moodle.com/branded-app/>

Pro	Premium
€199 EUR	€499 EUR
Price per year. Free if hosted by Moodle Certified Partner, or MoodleCloud.	Price per year. Free if hosted by Moodle Certified Partner, or MoodleCloud.
500 active devices for push notifications 4 offline courses	- Unlimited active devices for push notifications - Unlimited offline courses
What's included?	What's included?
Sign up	Sign up

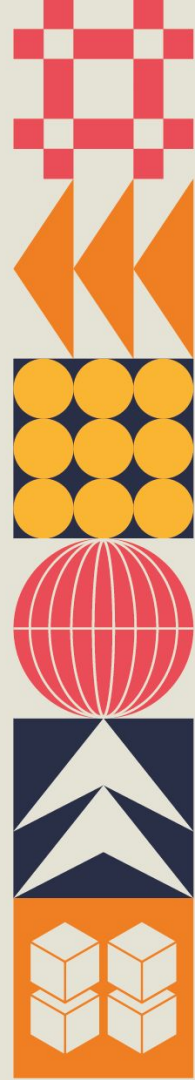
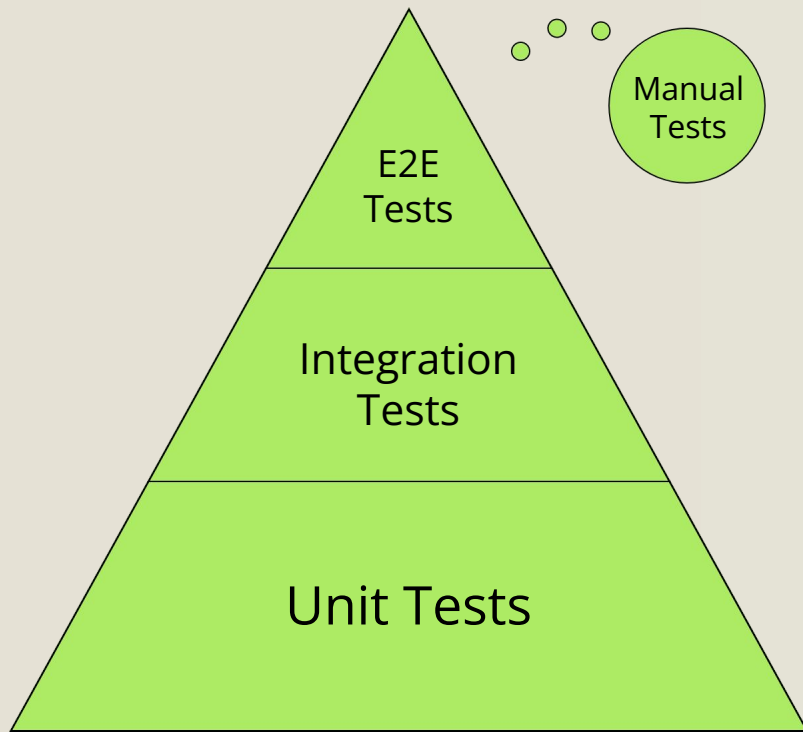




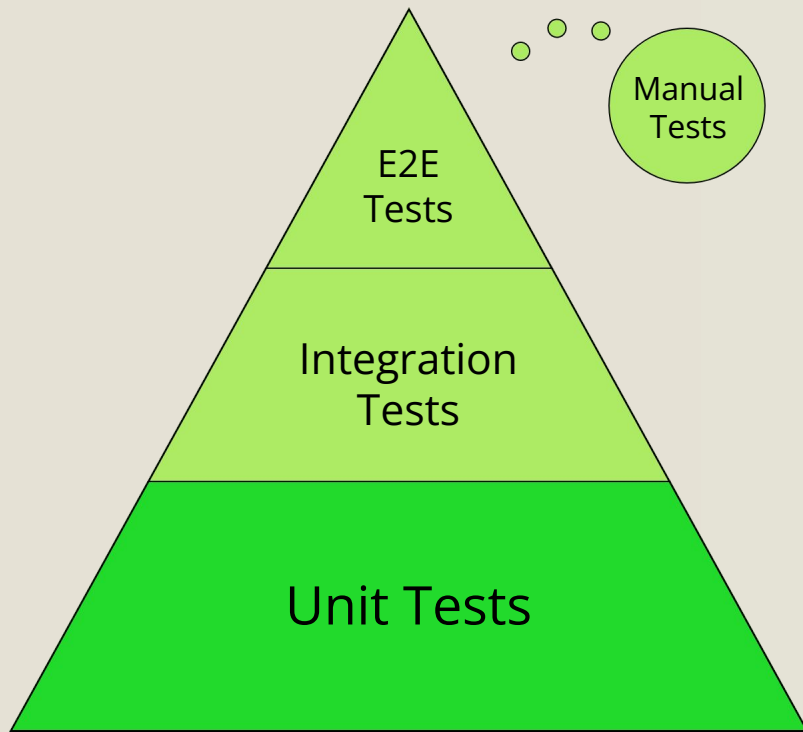
Testing



What is “Testing”?

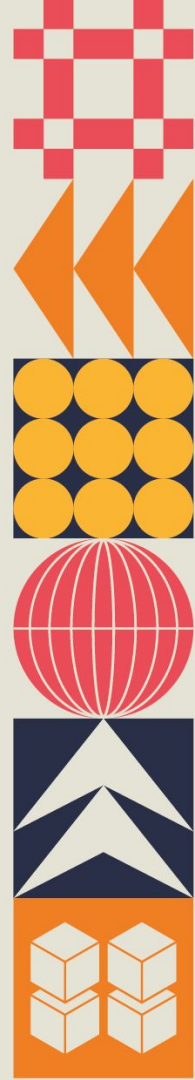


What is “Testing”?

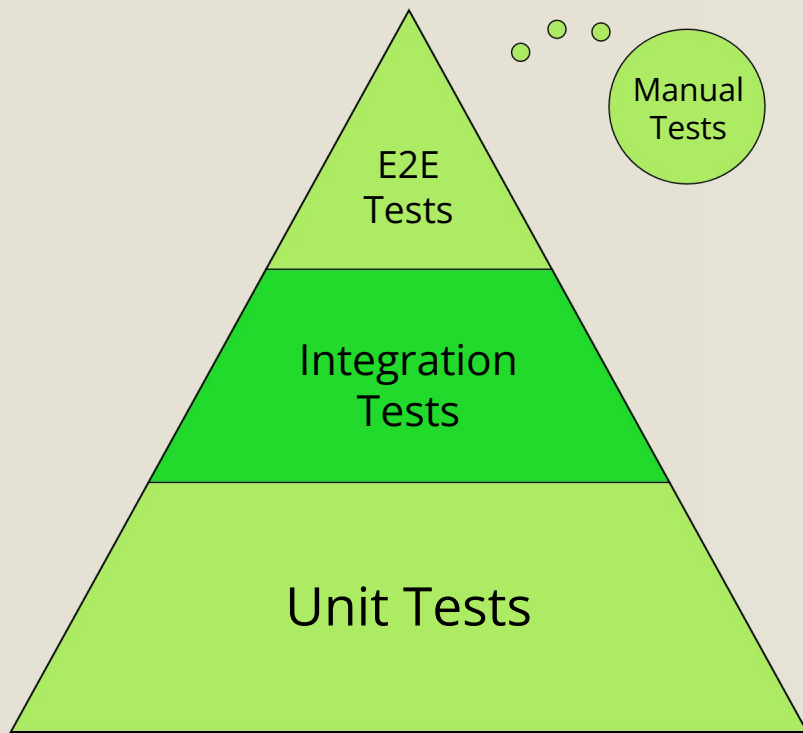


Unit Tests

- Test units in isolation
- Useful to cover edge cases
- Usually very fast

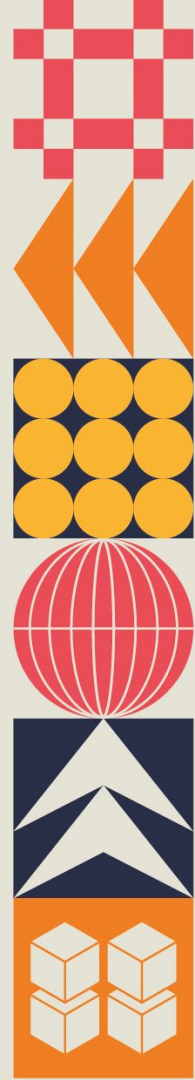


What is “Testing”?

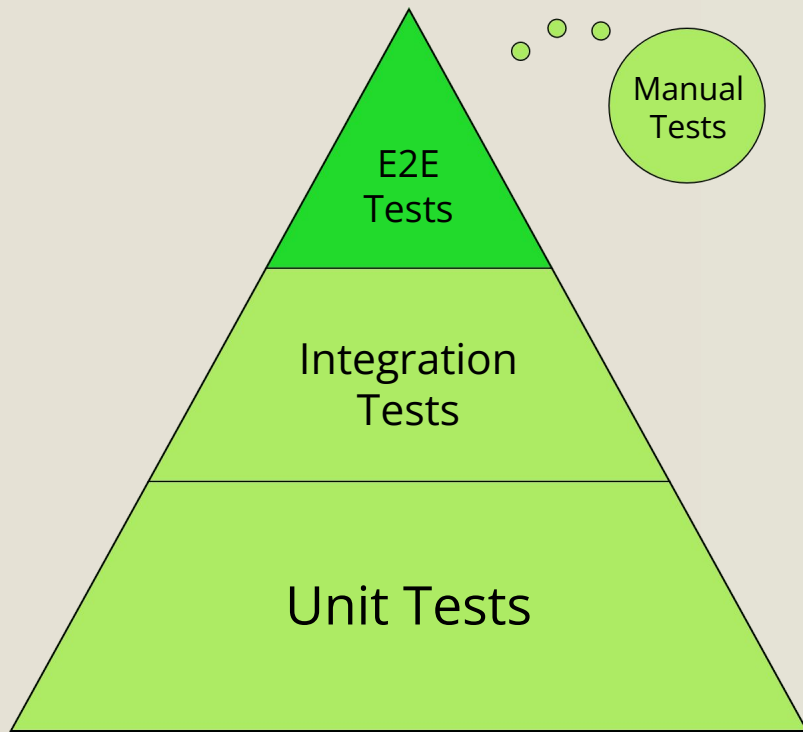


Integration Tests

- Test units working together
- Useful to cover complex interactions

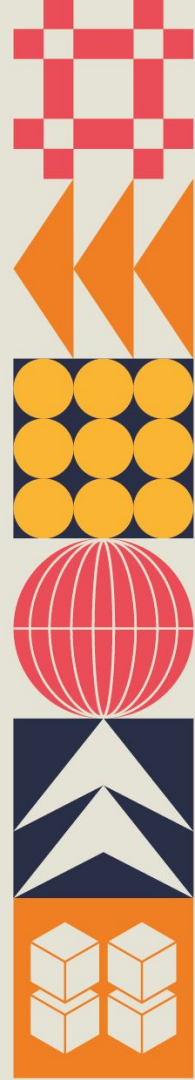


What is “Testing”?

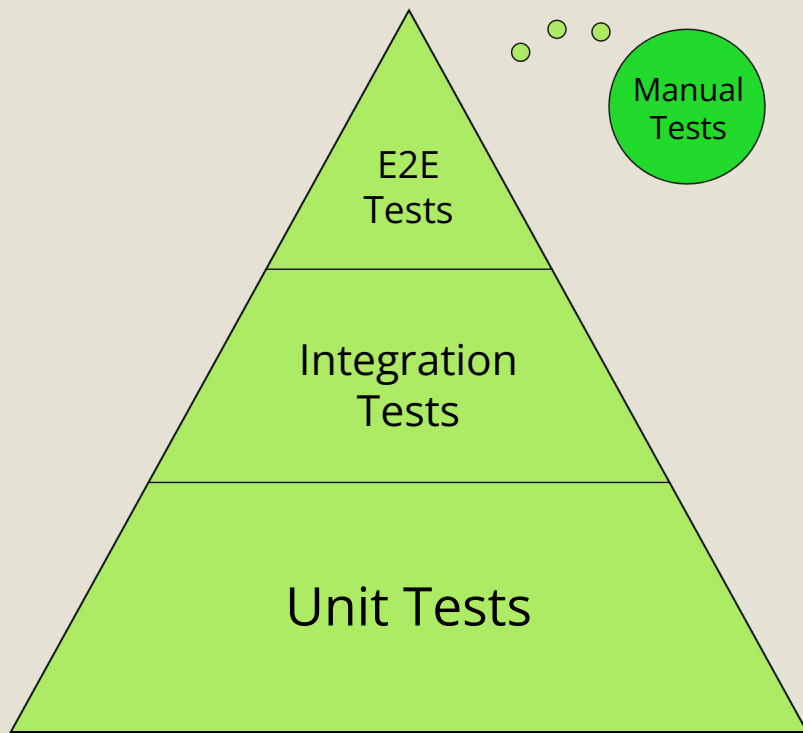


End-to-end (E2E) Tests

- Test systems working together
- As close to production as possible
- Useful to cover golden paths
- Usually slow

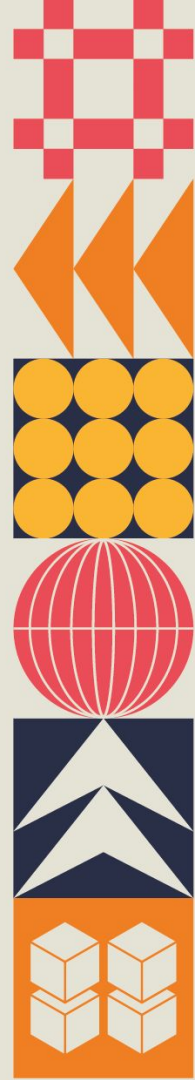


What is “Testing”?



Manual Tests

- Tests performed without automation
- Usually the most expensive, require human labor

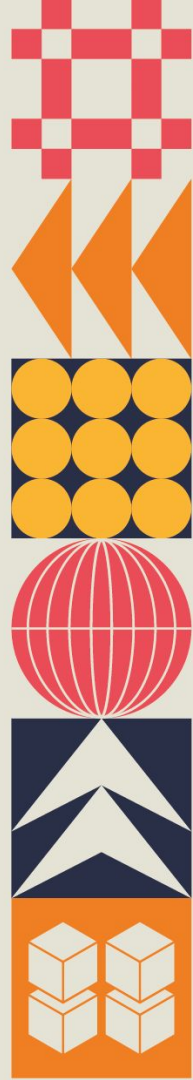


Why Test? 5 reasons



1. Peace of Mind

A well-tested codebase
lets you sleep at night.



Why Test? 5 reasons



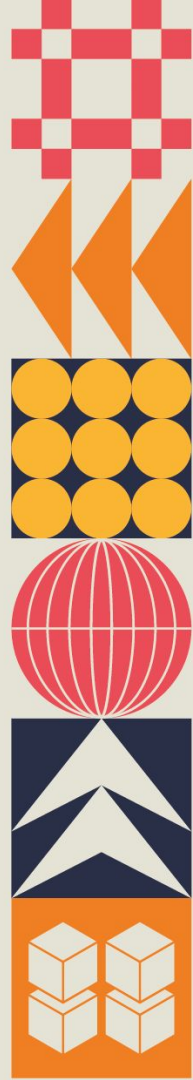
1. Peace of Mind

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2. Avoid Regressions

Tests enable safer
refactors.



Why Test? 5 reasons



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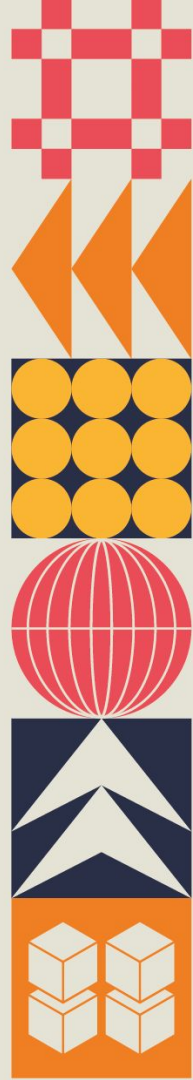
3. Reduce Bugs

Fix a bug once and it's fixed
for now, write a test and
it's fixed forever.



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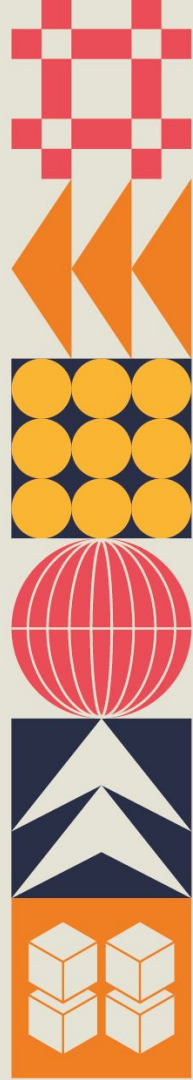
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4. Handshake Overhead

Make sure that new
features don't break
existing functionality.



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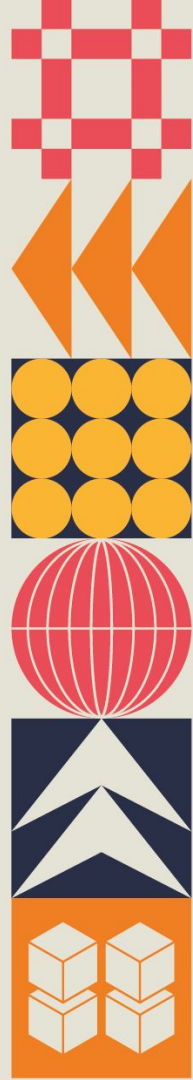
4. Handshake Overhead

Make sure that new
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existing functionality.



5. DRY

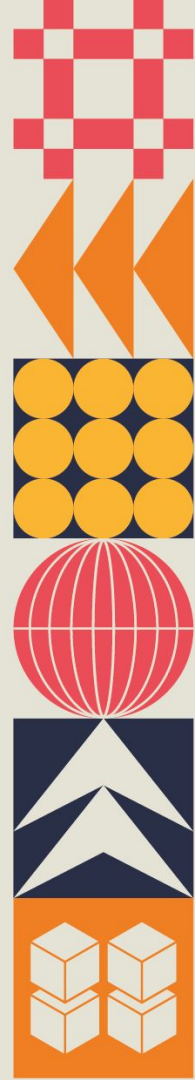
You are already testing,
you may as well
automate it.



Testing the Moodle App

Challenges

- Support old devices
(Android 5.1+, iOS 11+)
- Support old LMS versions
(LMS 3.5+)
- Offline functionality
- Media playback



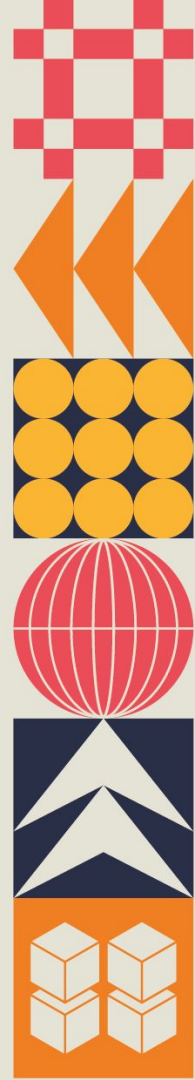
Testing the Moodle App

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Advantages

- Reduced functionality
(Mostly for students, some for teachers)
- Global updates



Testing the Moodle App

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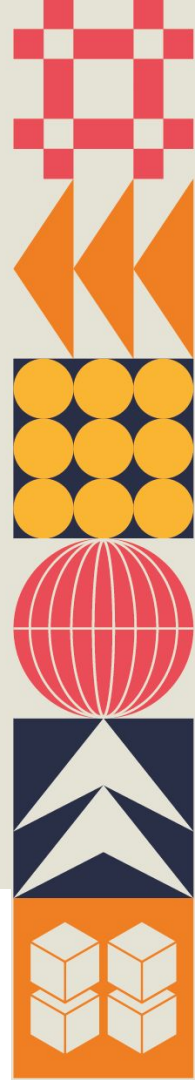
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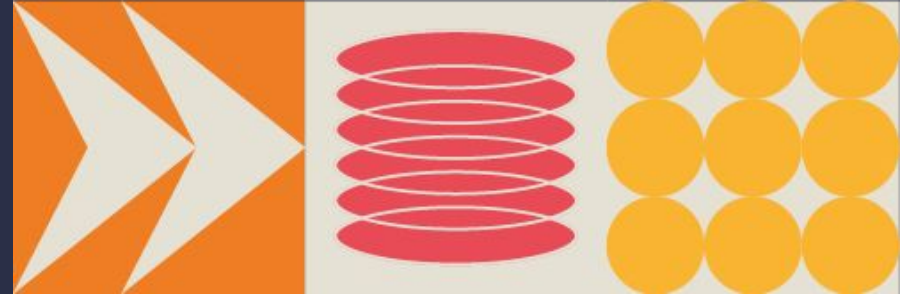
(Android 3.9.3 release, November 2020)

Installs targeted by rollout

6,697,023

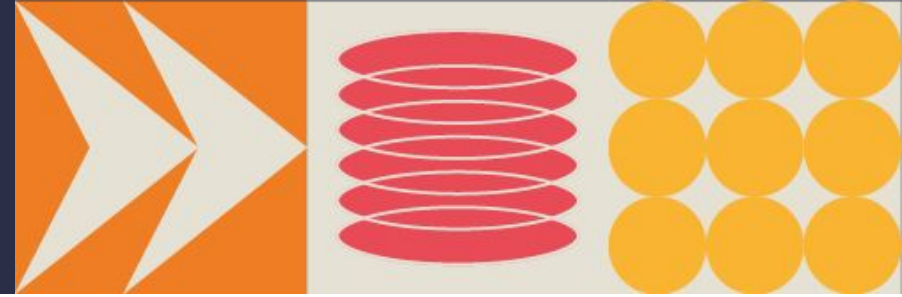


A Brief History of Testing the Moodle App



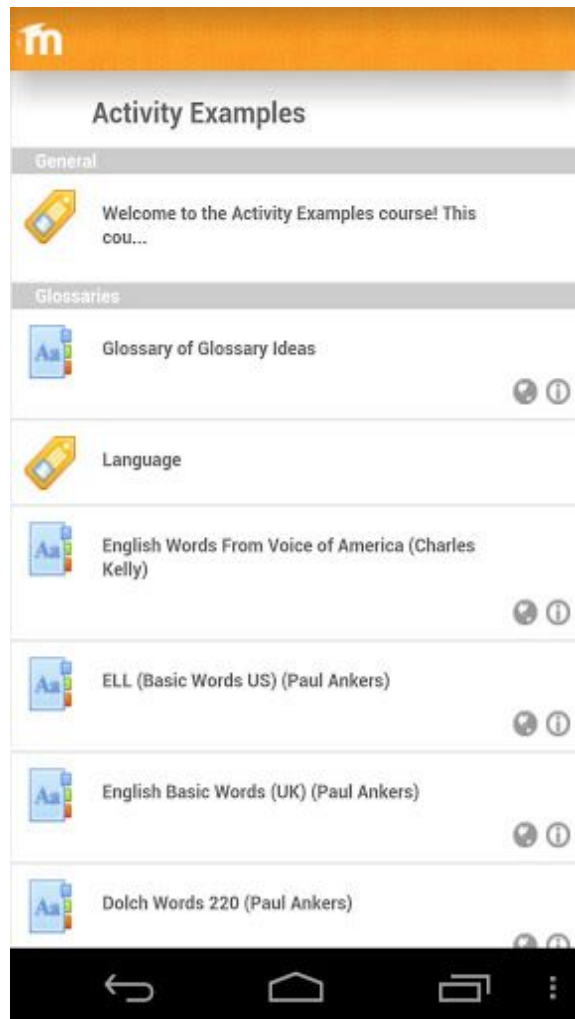
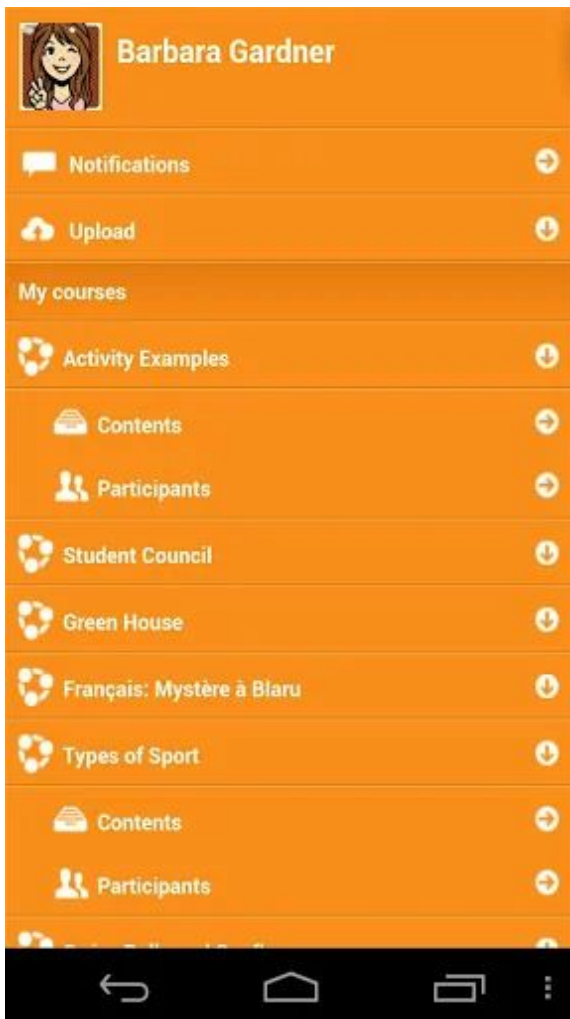
Initial Release

April 2013



- Initial release based on “Unofficial Moodle Mobile”
[MOBILE-153](#)

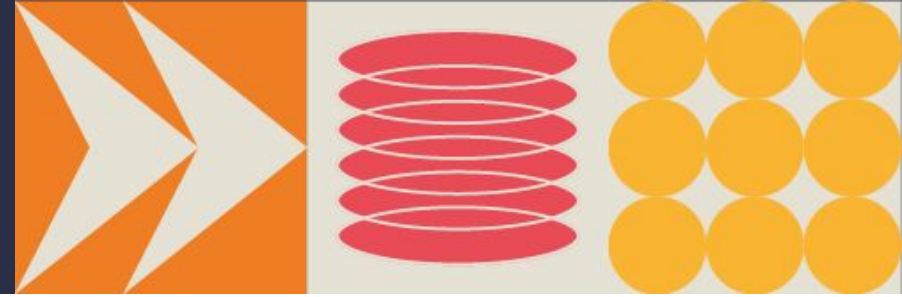




Team size: 1
Total tests: 0

Manual Tests

July 2016



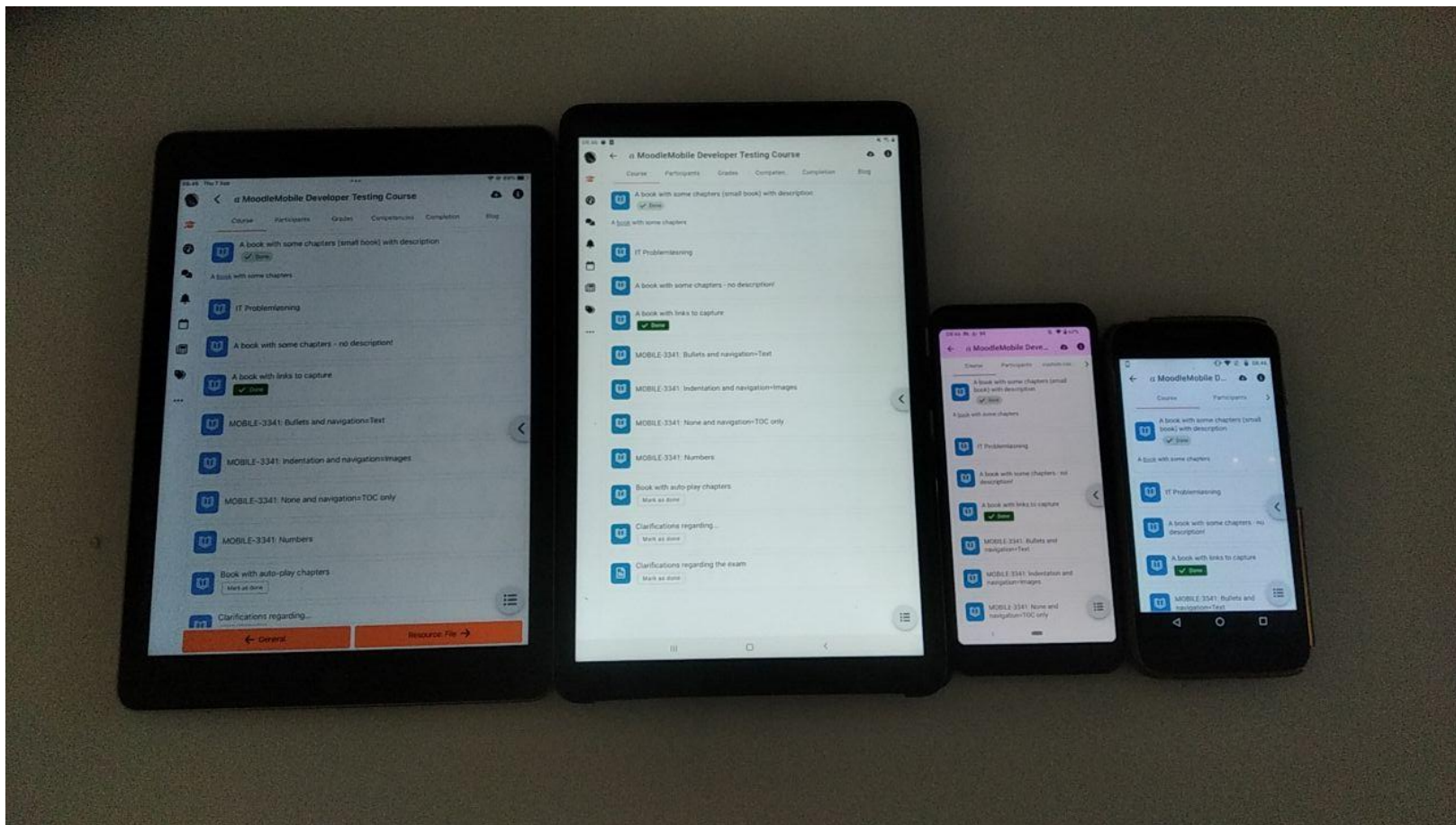
- App version 3.1.2
- Started with 156 tests



	A	B	C	D	E	F	G	H	I	J	K	M	N	O	P	Q	R	S	T	U	V
1		Test			Android								iOS								
2	Test case	On	Crit	News in	2.6	2.7	2.8	2.9	3.0	3.1	3.1NL	2.6	2.7	2.8	2.9	3.0	3.1	3.1NL	Fail	Worked	Notes
3	Login																		26	151	
4	Add a new site login inapp	x								2	1						1	1	0	4	
5	Add a new site login SSO via new browser window				1								1						0	2	
6	Add a new site login SSO via embeded browser					1								1					0	2	
7	Delete a site	x								2	1						1	1	0	4	
8	Login with demo account										1							1	0	1	
9	Reconnect (token expired)	x		X						1	1						1	1	0	3	
10	Fixed siteurl (custom app)																		0	0	
11	My Courses																				
12	See my courses									2	1						1	1	0	4	
13	Filter courses									2	1						1	1	0	4	
14	Search for a course									2	1						1	1	0	4	
15	View course contents									2	1						1	1	0	4	
16	View section contents									2	1						1	1	0	4	
17	Self enrol									2	1						1	1	0	4	
18	Guest access									2	1						1	1	0	4	
19	Activity completion									2	1						1	1	0	4	
20	Prefetch sections		x							-2	1						-1	1	-3	1	
21	Pull to refresh									2	1						1	1	0	4	
22	Calendar events																				
23	List events by date									2	1						1	1	0	4	
24	Local notifications									2	1						1	1	0	4	
25	Click local notification		x							1	1						-1	1	-1	2	Kill the app, click notif, event was opened. I clicked back and the event was opened aga
26	Competencies																				
27	View site learning plans, course and user competencies									2							1		0	3	
28	View evidences									2							1		0	3	
29	View certain user competencies in course									1							1		0	2	
30	Course completion																				
31	View own course completion (if enabled)									2	1						1	1	0	4	
32	View certain user course completion									1	1						1	1	0	3	
33	Manual self completion									2	1						1	1	0	4	
34	Files																				
35	View site files and private files									2	1		-1				1	1	-1	4	In 2.6 files couldn't be viewed because mimetype is always application/x-forcedownload

Toggle passed

Title	Sites	Role/User	Tests	Android	iOS	Issue
Filter by name or status... ?		958/958 (100%)		478/478 (100%)	480/480 (100%)	
Login Android assignee: Juan Leyva iOS assignee: Juan Leyva						
		55/55 (100%)		28/28 (100%)	27/27 (100%)	
1: Add a new site in the app	master					
2: Find site by name						
- Find the Learn Moodle site by searching: Learn Moodle MOOC - Find the Learn Moodle site by searching: learn.moodle.org - Confirm that a custom image is displayed						
5: Login SSO via new browser window core preferences	3.5					
6: Login SSO via embeded browser core preferences	3.5					
7: Add a non existing site						
8: Add a non mobile WS enabled site						
9: Add a site and login with wrong user/password	master					



Behat Steps

February 2019



- Behat steps for the Moodle App integrated in core - [MDL-63977](#)
- Community contribution
(Thanks to Sam Marshall from the Open University!)



@app @javascript

Feature: Test app (demo)

In order to test something in the app

As a developer

I need for this test script to run the app

Background:

Given the following "courses" exist:

fullname	shortname
Course 1	C1

And the following "users" exist:

username
student1

And the following "course enrolments" exist:

user	course	role
student1	C1	student

Scenario: Try going into the course

When I enter the app

And I log in as "student1"

And I press "Course 1" near "Course overview" in the app

Then the header should be "Course 1" in the app

Behat Tests

June 2020



- Behat tests implemented in the app - [MOBILE-3294](#)
- Have changed a lot since: [MOBILE-3738](#), [MDL-68969](#), [MOBILE-4110](#)
- Read the docs: [Acceptance testing for the Moodle App](#)





All checks have passed

2 successful checks

Hide all checks




Acceptance tests (Behat) / behat (pull_request)

Successful in 69m

Details




Testing / test (pull_request)

Successful in 10m

Required

Details

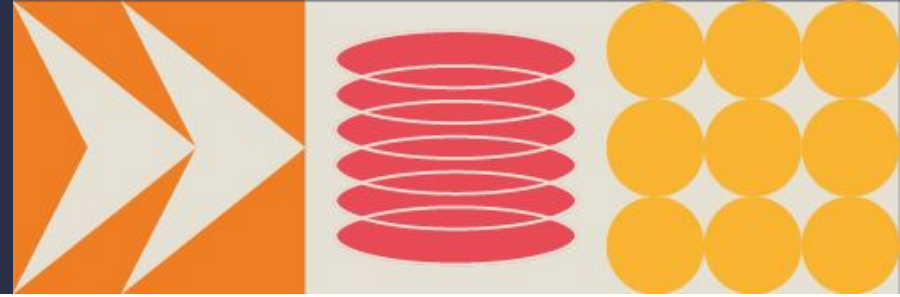
S	W	Name ↓	Last Success	Last Failure	Last Duration	Build description
		MMP.01.01 - Behat - Chrome + Stable App	20 hr #548	14 days #536	46 min	
		MMP.02.01 - Behat - Chrome + Dev App	13 hr #537	4 days 13 hr #534	44 min	
		MMP309.01.01 - Behat - Chrome + Stable App	3 days 2 hr #258	2 mo 24 days #229	32 min	
		MMP309.02.01 - Behat - Chrome + Dev App	2 days 15 hr #390	1 mo 4 days #375	36 min	
		MMP311.01.01 - Behat - Chrome + Stable App	3 days 5 hr #371	2 mo 24 days #342	40 min	
		MMP311.02.01 - Behat - Chrome + Dev App	2 days 18 hr #435	9 days 18 hr #431	37 min	
		MMP400.01.01 - Behat - Chrome + Stable App	2 days 16 hr #253	2 mo 26 days #217	46 min	
		MMP400.02.01 - Behat - Chrome + Dev App	3 days 3 hr #261	1 mo 4 days #245	44 min	
		MMP401.01.01 - Behat - Chrome + Stable App	5 hr 40 min #105	6 days 5 hr #101	39 min	
		MMP401.02.01 - Behat - Chrome + Dev App	2 hr 20 min #107	1 mo 4 days #81	45 min	
		MMP402.01.01 - Behat - Chrome + Stable App	4 hr 50 min #91	12 days #82	37 min	
		MMP402.02.01 - Behat - Chrome + Dev App	10 hr #98	4 days 10 hr #95	47 min	

Icon: S M L

Icon legend  Atom feed for all  Atom feed for failures  Atom feed for just latest builds

Unit Tests

November 2020



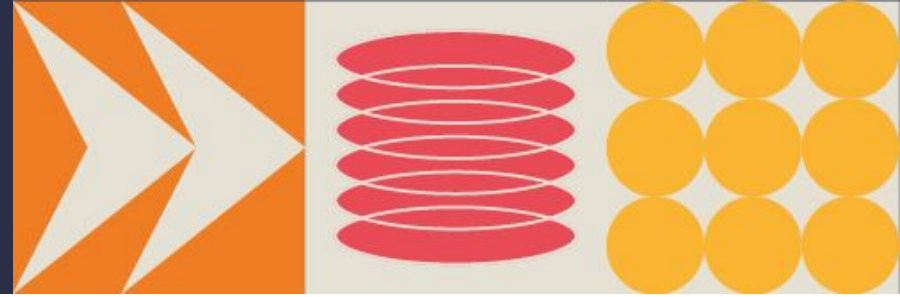
- Added Unit tests (using Jest) - [MOBILE-3566](#)
- Read the docs: [Unit testing for the Moodle App](#)



```
it('guesses the Moodle domain', () => {  
  // Check known endpoints first.  
  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/my')).toEqual('school.edu/moodle');  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/?redirect=0')).toEqual('school.edu/moodle');  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/index.php')).toEqual('school.edu/moodle');  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/course/view.php')).toEqual('school.edu/moodle');  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/login/index.php')).toEqual('school.edu/moodle');  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/mod/page/view.php')).toEqual('school.edu/moodle');  
  
  // Check an unknown endpoint.  
  
  expect(CoreUrl.guessMoodleDomain('https://school.edu/moodle/unknown/endpoint.php')).toEqual('school.edu');  
});
```

Performance Tests

September 2021



- Added performance tests - [MOBILE-3875](#)
- Collects similar metrics to Lighthouse
<https://github.com/GoogleChrome/lighthouse>



(index)	Total duration	Scripting	Styling	Blocking	# Network requests	# DB Queries	# Long Tasks	# runs
First Contentful Paint	'2367ms'	'804ms'	'76ms'	'0ms'	5	59	0	3
Login	'7779ms'	'1181ms'	'137ms'	'0ms'	9	155	0	3
Open Activity	'7134ms'	'2247ms'	'299ms'	'0ms'	32	85	1	3
Time to Interactive	'3641ms'	'809ms'	'76ms'	'1ms'	1	60	0	3
Total Blocking Time	'1351ms'	'651ms'	'31ms'	'0ms'	0	1	0	3

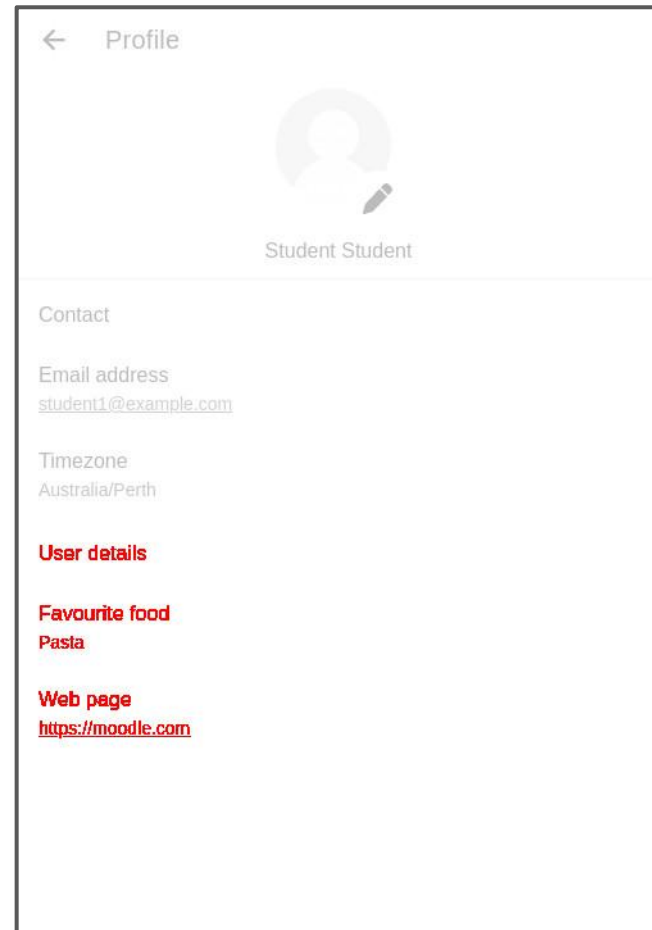
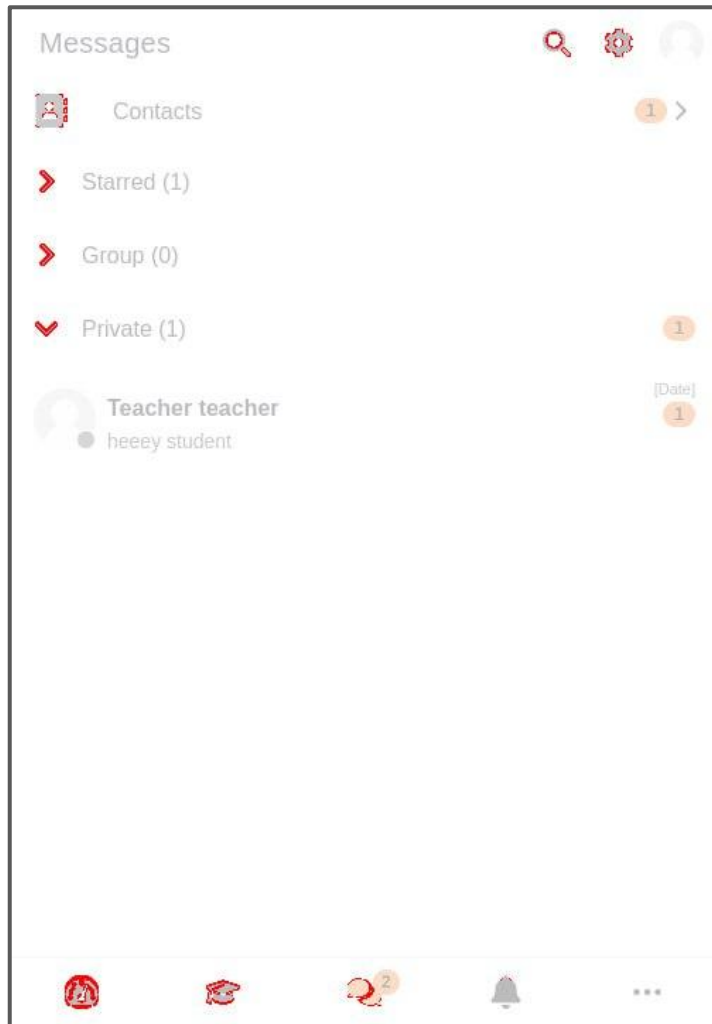
Snapshot Tests

February 2023



- Added Behat snapshot tests - [MOBILE-4254](#)
- Idea born in the 2022 Moot's Moodle Stand!
- Can also be used in the LMS: [local_behatsnapshots](#)





Deleted

← Profile



Student Student

Contact

Email address

student1@example.com

Timezone

Australia/Perth

W: 500px | H: 720px

Added

← Profile



Student Student

Contact

Email address

student1@example.com

Timezone

Australia/Perth

User details

Favourite food

Pasta

Web page

<https://moodle.com>

W: 500px | H: 720px

Tracker Integration

July 2023



- Added app tests to the tracker
- Still fine-tuning it
 - App tests are flaky sometimes.
 - They may run on older commits before rebasing.
 - Some app tests are coupled to LMS UI (lack of generators).

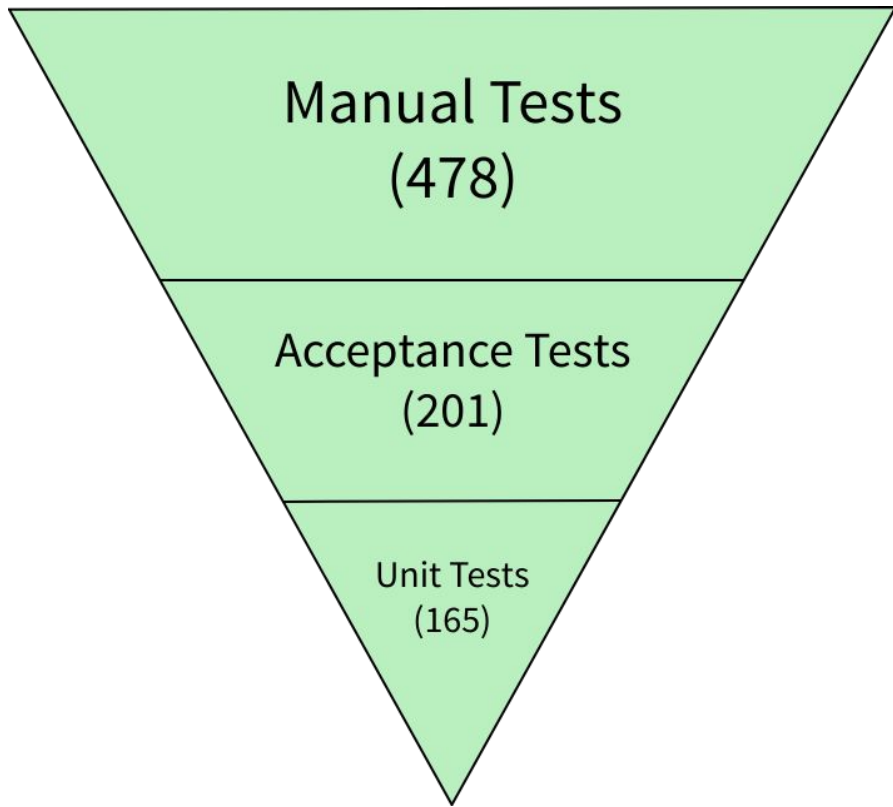


Timeline Recap



Testing Today

- Inverted pyramid?
- Improved type-safety with TypeScript
 - Removed “any”
 - Typed webservice responses
 - Branded types
 - Typed routes POC
 - Moving towards strict types
 - ...
- Team size: 15
(1 dedicated QA person)
- You can help us!
Join the beta testing in Google Play for Android
and TestFlight for iOS





moodlemoot.org | [#MootGlobal23](https://twitter.com/MootGlobal23)

E: noel@moodle.com

